In [1]:
```
import numpy as np
```

In [2]:
```
print(np.__version__)
```

```
1.21.5
```

In [3]:
```
x=10
print(x,type(x))
```

```
10 <class 'int'>
```

In [5]:
```
a=np.array(x)
print(a,type(a))
```

```
10 <class 'numpy.ndarray'>
```

In [6]:
```
a
```

Out[6]:
```
array(10)
```

In [7]:
```
a.ndim
```

Out[7]:
```
0
```

In [8]:
```
a.shape
```

Out[8]:
```
()
```

In [9]:
```
x=[10]
print(x,type(x))
```

```
[10] <class 'list'>
```

In [10]:
```
a=np.array(x)
print(a,type(a))
```

```
[10] <class 'numpy.ndarray'>
```

In [11]:
```
a
```

Out[11]:
```
array([10])
```

In [12]:
```
a.ndim
```

Out[12]:
```
1
```

In [13]:
```
a.shape
```

Out[13]:
```
(1,)
```

In [14]:
```
r=range(10,25)
print(r,type(r))
```

```
range(10, 25) <class 'range'>
```

In [16]:
```
a=np.array(r)
print(a,type(a),len(a))
```

```
[10 11 12 13 14 15 16 17 18 19 20 21 22 23 24] <class 'numpy.ndarray'> 15
```

In [17]:
```
a.ndim
```

Out[17]:
```
1
```

In [18]:
```
a.shape
```

Out[18]:
```
(15,)
```

In [19]:
```
a.shape=(3,5)
```

In [20]:
```
print(a,type(a))
```

```
[[10 11 12 13 14]
 [15 16 17 18 19]
 [20 21 22 23 24]] <class 'numpy.ndarray'>
```

In [21]:
```
a.ndim
```

Out[21]:
```
2
```

In [22]:
```
a.shape
```

Out[22]:
```
(3, 5)
```

In [23]:
```
a.shape=(5,3)
```

In [24]:
```
a
```

Out[24]:
```
array([[10, 11, 12],
       [13, 14, 15],
       [16, 17, 18],
       [19, 20, 21],
       [22, 23, 24]])
```

In [25]:
```
a.ndim
```

Out[25]:
```
2
```

In [26]:
```
a.shape
```

Out[26]:
```
(5, 3)
```

In [27]:
```
r=range(10,25)
print(r,type(r))
a=np.array(r,dtype=float)
print(a,type(a),len(a))
```

```
range(10, 25) <class 'range'>
[10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24.] <class 'numpy.ndarray'> 1
5
```

In [28]:
```
a.dtype
```

Out[28]:
```
dtype('float64')
```

In [29]:
```
r=range(10,25)
print(r,type(r))
a=np.array(r,)
print(a,type(a),len(a))
```

```
range(10, 25) <class 'range'>
[10 11 12 13 14 15 16 17 18 19 20 21 22 23 24] <class 'numpy.ndarray'> 15
```

In [30]:
```
a.dtype
```

Out[30]:
```
dtype('int32')
```

In [31]:
```
lst=[10,20,35,67,12,34]
print(lst,type(lst))
a=np.array(lst)
print(a,type(a))
```

```
[10, 20, 35, 67, 12, 34] <class 'list'>
[10 20 35 67 12 34] <class 'numpy.ndarray'>
```

In [34]:
```
print("Dimesion=",a.ndim)
print("shape=",a.shape)
print("Data Type=",a.dtype)
```

```
Dimesion= 1
shape= (6,)
Data Type= int32
```

In [35]:
```
a.shape=(3,2)
```

In [36]:
```
print(a)
print("Dimesion=",a.ndim)
print("shape=",a.shape)
print("Data Type=",a.dtype)
```

```
[[10 20]
 [35 67]
 [12 34]]
Dimesion= 2
shape= (3, 2)
Data Type= int32
```

In [37]:
```
a.shape=(2,3)
print(a)
print("Dimesion=",a.ndim)
print("shape=",a.shape)
print("Data Type=",a.dtype)
```

```
[[10 20 35]
 [67 12 34]]
Dimesion= 2
shape= (2, 3)
Data Type= int32
```

In [38]:
```
tpl=(10,20,35,67,12,34)
print(tpl,type(tpl))
a=np.array(tpl)
print(a,type(a))
```

```
(10, 20, 35, 67, 12, 34) <class 'tuple'>
[10 20 35 67 12 34] <class 'numpy.ndarray'>
```

In [39]:
```
a.shape=(2,3)
print(a)
print("Dimesion=",a.ndim)
print("shape=",a.shape)
print("Data Type=",a.dtype)
```

```
[[10 20 35]
 [67 12 34]]
Dimesion= 2
shape= (2, 3)
Data Type= int32
```

In [40]:
```
s={10,20,35,67,12,34}
print(s,type(s))
a=np.array(s)
print(a,type(a))
```

```
{34, 35, 67, 20, 10, 12} <class 'set'>
{34, 35, 67, 20, 10, 12} <class 'numpy.ndarray'>
```

In [41]:
```
print(a.ndim)
print(a.shape)
```

```
0
()
```

In [42]:
```
a
```

Out[42]:
```
array({34, 35, 67, 20, 10, 12}, dtype=object)
```

In [43]:
```
d1={10:"Apple",20:"Mango",30:"Kiwi",40:"Guava"}
print(d1,type(d1))
```

```
{10: 'Apple', 20: 'Mango', 30: 'Kiwi', 40: 'Guava'} <class 'dict'>
```

In [44]:
```
a=np.array(d1)
print(a,type(a))
```

```
{10: 'Apple', 20: 'Mango', 30: 'Kiwi', 40: 'Guava'} <class 'numpy.ndarray'>
```

In [45]:
```
a
```

Out[45]:
```
array({10: 'Apple', 20: 'Mango', 30: 'Kiwi', 40: 'Guava'}, dtype=object)
```

In [46]:
```
print(a.ndim)
print(a.shape)
```

```
0
()
```

In [47]:
```
lst=[10,20,30,40,50,60,70,80,90,15,25,35,55,65,25,75]
print(lst,type(lst))
```

```
[10, 20, 30, 40, 50, 60, 70, 80, 90, 15, 25, 35, 55, 65, 25, 75] <class 'list'>
```

In [48]:
```
a=np.array(lst)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
print("Data Type=",a.dtype)
```

```
[10 20 30 40 50 60 70 80 90 15 25 35 55 65 25 75] <class 'numpy.ndarray'>
Dimesion= 1
shape= (16,)
Data Type= int32
```

In [49]:
```
a.shape=(4,4)
print(a)
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[10 20 30 40]
 [50 60 70 80]
 [90 15 25 35]
 [55 65 25 75]]
Dimesion= 2
shape= (4, 4)
```

In [50]:
```
a.shape=(8,2)
print(a)
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[10 20]
 [30 40]
 [50 60]
 [70 80]
 [90 15]
 [25 35]
 [55 65]
 [25 75]]
Dimesion= 2
shape= (8, 2)
```

In [51]:
```
a.shape=(2,4,2)
print(a)
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[[10 20]
  [30 40]
  [50 60]
  [70 80]]

 [[90 15]
  [25 35]
  [55 65]
  [25 75]]]
Dimesion= 3
shape= (2, 4, 2)
```

In [52]:
```
a.shape=(2,2,2,2)
print(a)
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[[[10 20]
   [30 40]]

  [[50 60]
   [70 80]]]


 [[[90 15]
   [25 35]]

  [[55 65]
   [25 75]]]]
Dimesion= 4
shape= (2, 2, 2, 2)
```

In [53]:
```
a.shape=(16,)
print(a)
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[10 20 30 40 50 60 70 80 90 15 25 35 55 65 25 75]
Dimesion= 1
shape= (16,)
```

In [54]:
```
Function Name---arange()
Syntax1: numpy.arnage(Value)-----> 0 to Value-1
Syntax2: numpy.arnage(Begin,End)-----> Begin to End-1
Syntax3: numpy.arnage(Begin,End,Step)-----> Begin to End-1 with equal Interval with step
```

In [55]:
```
lst=[[10,20,30,40],[50,60,70,80],[90,15,25,35],[55,65,25,75]]
print(lst,type(lst))
a=np.array(lst)
print(a)
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[10, 20, 30, 40], [50, 60, 70, 80], [90, 15, 25, 35], [55, 65, 25, 75]] <class 'list'>
[[10 20 30 40]
 [50 60 70 80]
 [90 15 25 35]
 [55 65 25 75]]
Dimesion= 2
shape= (4, 4)
```

In [56]:
```
Function Name---arange()
Syntax1: numpy.arnage(Value)-----> 0 to Value-1
Syntax2: numpy.arnage(Begin,End)-----> Begin to End-1
Syntax3: numpy.arnage(Begin,End,Step)-----> Begin to End-1 with equal Interval with step
```

In [58]:
```
a=np.arange(9)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[0 1 2 3 4 5 6 7 8] <class 'numpy.ndarray'>
Dimesion= 1
shape= (9,)
```

In [59]:
```
a.shape=(3,3)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[0 1 2]
 [3 4 5]
 [6 7 8]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (3, 3)
```

In [60]:
```
a=np.arange(10,35)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33
 34] <class 'numpy.ndarray'>
Dimesion= 1
shape= (25,)
```

In [61]:
```
a.shape=(5,5)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[10 11 12 13 14]
 [15 16 17 18 19]
 [20 21 22 23 24]
 [25 26 27 28 29]
 [30 31 32 33 34]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (5, 5)
```

In [62]:
```
a=np.arange(10,51,5)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[10 15 20 25 30 35 40 45 50] <class 'numpy.ndarray'>
Dimesion= 1
shape= (9,)
```

In [63]:
```
a.shape=(3,3)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[10 15 20]
 [25 30 35]
 [40 45 50]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (3, 3)
```

In [64]:
```
#Function  Name:  zeros()
#>Syntax: varname=numpy.zeros(shape,dtype)
```

In [65]:
```
a=np.zeros(6)
print(a,type(a))
```

```
[0. 0. 0. 0. 0. 0.] <class 'numpy.ndarray'>
```

In [66]:
```
a=np.zeros(6,dtype=int)
print(a,type(a))
```

```
[0 0 0 0 0 0] <class 'numpy.ndarray'>
```

In [67]:
```
a.shape=(3,2)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[0 0]
 [0 0]
 [0 0]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (3, 2)
```

In [68]:
```
a=np.zeros((3,3),int)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[0 0 0]
 [0 0 0]
 [0 0 0]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (3, 3)
```

In [69]:
```
a=np.zeros((4,5),int)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[0 0 0 0 0]
 [0 0 0 0 0]
 [0 0 0 0 0]
 [0 0 0 0 0]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (4, 5)
```

In [70]:
```
a=np.zeros((2,4,5),int)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[[0 0 0 0 0]
  [0 0 0 0 0]
  [0 0 0 0 0]
  [0 0 0 0 0]]

 [[0 0 0 0 0]
  [0 0 0 0 0]
  [0 0 0 0 0]
  [0 0 0 0 0]]] <class 'numpy.ndarray'>
Dimesion= 3
shape= (2, 4, 5)
```

In [71]:
```
#Function  Name:  ones()
#>Syntax: varname=numpy.ones(shape,dtype)
```

In [72]:
```
a=np.ones(8)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[1. 1. 1. 1. 1. 1. 1. 1.] <class 'numpy.ndarray'>
Dimesion= 1
shape= (8,)
```

In [73]:
```
a=np.ones(8,int)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[1 1 1 1 1 1 1 1] <class 'numpy.ndarray'>
Dimesion= 1
shape= (8,)
```

In [74]:
```
a.shape=(4,2)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[1 1]
 [1 1]
 [1 1]
 [1 1]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (4, 2)
```

In [75]:
```
a.shape=(2,2,2)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[[1 1]
  [1 1]]

 [[1 1]
  [1 1]]] <class 'numpy.ndarray'>
Dimesion= 3
shape= (2, 2, 2)
```

In [76]:
```
a=np.ones((4,3),int)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[1 1 1]
 [1 1 1]
 [1 1 1]
 [1 1 1]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (4, 3)
```

In [77]:
```
a=np.ones((3,4,3),int)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[[1 1 1]
  [1 1 1]
  [1 1 1]
  [1 1 1]]

 [[1 1 1]
  [1 1 1]
  [1 1 1]
  [1 1 1]]

 [[1 1 1]
  [1 1 1]
  [1 1 1]
  [1 1 1]]] <class 'numpy.ndarray'>
Dimesion= 3
shape= (3, 4, 3)
```

In [78]:
```
#Function  Name:  full()
#>Syntax: varname=numpy.full(shape,fill_value,dtype)
```

In [82]:
```
a=np.full(4,fill_value=8)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[8 8 8 8] <class 'numpy.ndarray'>
Dimesion= 1
shape= (4,)
```

In [83]:
```
a.shape=(2,2)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[8 8]
 [8 8]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (2, 2)
```

In [84]:
```
a=np.full((4,4),fill_value="KVR")
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[['KVR' 'KVR' 'KVR' 'KVR']
 ['KVR' 'KVR' 'KVR' 'KVR']
 ['KVR' 'KVR' 'KVR' 'KVR']
 ['KVR' 'KVR' 'KVR' 'KVR']] <class 'numpy.ndarray'>
Dimesion= 2
shape= (4, 4)
```

In [85]:
```
a=np.full((4,4),fill_value=4)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[4 4 4 4]
 [4 4 4 4]
 [4 4 4 4]
 [4 4 4 4]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (4, 4)
```

In [86]:
```
a=np.full((2,4,4),fill_value=9)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[[9 9 9 9]
  [9 9 9 9]
  [9 9 9 9]
  [9 9 9 9]]

 [[9 9 9 9]
  [9 9 9 9]
  [9 9 9 9]
  [9 9 9 9]]] <class 'numpy.ndarray'>
Dimesion= 3
shape= (2, 4, 4)
```

In [87]:
```
#Function  Name:  identity()
#>Syntax: varname=numpy.identity(N,dtype)
```

In [88]:
```
a=np.identity(3,int)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[1 0 0]
 [0 1 0]
 [0 0 1]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (3, 3)
```

In [89]:
```
a=np.identity(4,int)
print(a,type(a))
print("Dimesion=",a.ndim)
print("shape=",a.shape)
```

```
[[1 0 0 0]
 [0 1 0 0]
 [0 0 1 0]
 [0 0 0 1]] <class 'numpy.ndarray'>
Dimesion= 2
shape= (4, 4)
```

In [90]:
```
print(a[0,0]*a[1,1]*a[2,2]*a[3,3])
```

```
1
```

In [91]:
```
#Function  Name:  hstack()
#>Syntax: varname=numpy.hstack((ndarryobj1,ndarryobj1))
```

In [94]:
```
a=np.array([[10,20,30],[40,50,60]])
print(a,type(a))
print(a.ndim)
print(a.shape)
```

```
[[10 20 30]
 [40 50 60]] <class 'numpy.ndarray'>
2
(2, 3)
```

In [96]:
```
b=np.array([[100,200],[400,500]])
print(b,type(b))
print(b.ndim)
print(b.shape)
```

```
[[100 200]
 [400 500]] <class 'numpy.ndarray'>
2
(2, 2)
```

In [97]:
```
c=np.hstack((a,b))
print(c,type(c))
print(c.ndim)
print(c.shape)
```

```
[[ 10  20  30 100 200]
 [ 40  50  60 400 500]] <class 'numpy.ndarray'>
2
(2, 5)
```

In [98]:
```
c=np.hstack((b,a))
print(c,type(c))
print(c.ndim)
print(c.shape)
```

```
[[100 200  10  20  30]
 [400 500  40  50  60]] <class 'numpy.ndarray'>
2
(2, 5)
```

In [99]:
```
a=np.array([[10,20,30],[40,50,60]])
print(a,type(a))
print(a.ndim)
print(a.shape)
```

```
[[10 20 30]
 [40 50 60]] <class 'numpy.ndarray'>
2
(2, 3)
```

In [100]:
```
b=np.array([[10,20,30],[40,50,60],[70,80,90]])
print(b,type(b))
print(b.ndim)
print(b.shape)
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]] <class 'numpy.ndarray'>
2
(3, 3)
```

In [101]:
```
c=np.vstack((a,b))
print(c,type(c))
print(c.ndim)
print(c.shape)
```

```
[[10 20 30]
 [40 50 60]
 [10 20 30]
 [40 50 60]
 [70 80 90]] <class 'numpy.ndarray'>
2
(5, 3)
```

In [102]:
```
c=np.vstack((b,a))
print(c,type(c))
print(c.ndim)
print(c.shape)
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]
 [10 20 30]
 [40 50 60]] <class 'numpy.ndarray'>
2
(5, 3)
```

In [ ]: